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PROJECT DOCUMENTATION

BILLING AND RECEIPT SYSTEM

Course: C Programming Lab (Mini Project)
Department: B.Sc. Computer Science | Section 'C' | Year I

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1. Introduction

Billing and Receipt management is a fundamental component of any commercial or service-oriented business. This project, developed as part of the C Programming Lab course, implements a console-based Billing and Receipt System that automates the process of invoice generation, payment collection, and receipt issuance.

1.1 What is Billing?

Billing refers to the systematic process by which businesses generate and deliver invoices to customers for goods delivered or services rendered. It involves:

- Gathering customer and product/service details
- Computing item totals, taxes, and applicable discounts
- Creating a formal invoice document
- Delivering the invoice through appropriate channels
- Monitoring payment status and following up on dues

1.2 What is a Receipt?

A receipt is an official acknowledgement issued to a customer confirming that payment has been received. Receipts serve multiple purposes including:

- Proof of purchase and payment for the customer
- Supporting document for warranty and return claims
- Evidence required for tax filing and auditing
- Record-keeping for both the business and the buyer

1.3 Importance of the System

Efficient billing and receipt practices are critical for maintaining healthy cash flow, enhancing customer satisfaction, and ensuring compliance with tax and business regulations. An automated system reduces human error, accelerates the payment cycle, and provides accurate financial records.

2. Objectives

The primary objectives of this project are outlined below:

1. Ensure timely and accurate invoicing for products or services delivered, helping businesses request payment efficiently and avoid errors.
2. Automate the process of creating, sending, and managing bills to save time, reduce manual workload, and improve overall productivity.
3. Maintain comprehensive records of customer transactions and monitor payment status for effective accounts receivable management.
4. Improve cash flow and financial stability by enabling prompt payment collection and accurate financial reporting.
5. Support compliance with tax laws and business regulations by including necessary details in bills and supporting audits or reviews.
6. Provide clear, formal proof that payment has been received from the customer, completing the transaction legally and financially.
7. Aid in accurate bookkeeping and record maintenance for both the business and the customer.
8. Serve as supporting documents for product returns, warranty claims, and resolving disputes.
9. Facilitate tax filing, audits, and compliance by offering evidence of all transactions.

3. Problem Statement

Manual billing processes are error-prone, time-consuming, and difficult to scale. Businesses face several challenges that this system aims to address:

3.1 Clear Communication

Invoices and bills must use clear, unambiguous language describing goods or services, prices, and payment terms to avoid misunderstandings between the business and the customer.

3.2 Prompt Error Correction

Mistakes in billing or receipts — such as incorrect amounts or duplicate charges — need to be quickly acknowledged and rectified by issuing corrected invoices or credit notes.

3.3 Active Investigation and Dialogue

When discrepancies or disputes arise, businesses must engage in honest, prompt communication with customers to understand their concerns and provide supporting documentation.

3.4 Automation and Standardisation

Automated billing software and standardised processes reduce human errors, improve accuracy, and speed up payment reconciliation.

3.5 Documentation

Detailed records of all billing, receipt, and dispute resolution actions must be maintained to support transparency and future audits.

3.6 Flexibility and Negotiation

Disputes should be resolved amicably by negotiating partial credits, discounts, or revised payment plans while maintaining a firm position on valid charges.

4. System Requirements

4.1 Functional Requirements

#	Feature	Description
1	Invoice Generation	Automatically create, customise, and deliver bills using predefined logic.
2	Payment Tracking	Track outstanding, partial, and full settlements per customer.
3	Customer Management	Maintain customer profiles including purchase history and contact info.
4	Reporting & Analytics	Generate financial reports and revenue trends for decision-making.
5	Integration	Interface with accounting, CRM, and payment gateway systems.
6	Multi-user Access	Support role-based permissions for staff, management, and customers.
7	Security & Compliance	Ensure data protection and regulatory compliance in all transactions.

4.2 Non-Functional Requirements

- Performance: Receipt generated within 1 second of payment confirmation.
- Reliability: Zero data loss during transaction recording.
- Usability: Simple console-based interface operable without prior training.
- Portability: Runs on any system with GCC compiler installed.
- Maintainability: Modular code structure for easy updates.

5. System Design

5.1 System Architecture

The system follows a modular console-based architecture with four primary components that handle input, processing, output, and data persistence respectively. Each module is independent yet interlinked through function calls and shared data structures.

5.2 Module Description

Module	Sub-module	Responsibility
Input Module	Order Entry	Captures Product ID, Quantity, Coupon Code, Customer ID, and Payment Type.
Processing Module	Billing Engine	Price retrieval, tax calculation, discount application, and total computation.
Output Module	Receipt Generator	Prints receipt to screen and saves digital invoice to file (receipt.txt).
File Handling	Database Management	SAVE new transactions, UPDATE inventory, RETRIEVE product prices and history.

5.3 Billing Logic / Algorithm

The core billing algorithm follows the steps below:

10. Start the program and display the main menu.
11. Capture customer details: name and mobile number.
12. Capture item details: name, unit price, and quantity for each item.
13. Calculate bill total: Item Price × Quantity, summed across all items.
14. Apply tax or discount: if subtotal > Rs 200, apply 10% GST; else apply 5% discount.
15. Display the total amount payable to the customer.
16. Receive payment method selection: Cash, Card, or UPI.
17. Process payment and verify sufficiency of funds (for cash).
18. Update inventory and mark transaction as complete.
19. Generate and print/save the receipt. End.

6. Implementation

6.1 Technology Stack

Component	Technology Used
Programming Language	C (ANSI C / C99)
Compiler	GCC (MinGW on Windows)
IDE / Editor	VS Code / Code::Blocks / Turbo C
Data Storage	File I/O (receipt.txt)
Libraries Used	stdio.h, string.h, stdlib.h, time.h

6.2 Key Data Structures

Two struct types are used to model the system data:

- Item struct — holds item name, unit price, quantity, and computed line total.
- Bill struct — holds customer details, an array of Item objects, calculated subtotal, discount, tax, grand total, payment type, amount paid, and change due.

6.3 Core Functions

Function	Purpose
get_customer_details()	Reads customer name and phone number from stdin.
add_items()	Iteratively collects item name, price, and quantity.
calculate_bill()	Computes subtotal, applies discount or GST, finds grand total.
display_summary()	Tabular display of all items and billing totals.
process_payment()	Handles Cash/Card/UPI payment and validates sufficiency.
print_receipt()	Formatted receipt printed to the console.
save_receipt()	Writes identical receipt to receipt.txt for record-keeping.
current_datetime()	Returns current date and time string using time.h.

6.4 Tax and Discount Logic

The billing engine applies different financial rules based on the subtotal value:

Condition	Rule Applied	Rate
Subtotal > Rs 200	GST (Tax)	10%
Subtotal <= Rs 200	Small Purchase Discount	5%

7. Source Code (Key Sections)

7.1 Bill Calculation Function

```
void calculate_bill(Bill *b) {

    b->subtotal = 0.0f;
    for (int i = 0; i < b->item_count; i++)
        b->subtotal += b->items[i].total;
    if (b->subtotal > 200.0f) {
        b->tax = b->subtotal * TAX_RATE;   // 10% GST
    } else {
        b->discount = b->subtotal * DISCOUNT_RATE; // 5% off
    }    b->grand_total = b->subtotal - b->discount + b->tax;}
```

8. Output Screens

8.1 Sample Receipt Output

The following is a sample output generated by the program for a test transaction:

```
==================================================
              PAYMENT RECEIPT
==================================================
Store   : KG SMART STORE
Date    : 16-03-2026  10:30:00
Name    : Rohit Kumar
--------------------------------------------------
Item                      Qty   Price    Amount
--------------------------------------------------
Rice                        2   45.00     90.00
Milk                        3   25.50     76.50
Bread                       1   35.00     35.00
--------------------------------------------------
Subtotal (Rs)                            201.50
GST 10% (Rs)                              20.15
==================================================
GRAND TOTAL (Rs)                         221.65
==================================================
Payment Method : Cash
Amount Paid    : Rs 300.00
Change Due     : Rs 78.35
** Thank you, Rohit Kumar! Visit again. **
==================================================
```

8.2 Output Screens Description

#	Screen	Description
1	Dashboard	Displays the main menu to start a new bill or exit.
2	Customer Entry	Prompts for customer name and mobile number.
3	Item Entry	Collects item name, price, and quantity iteratively.
4	Bill Summary	Tabular view of all items with subtotal, tax/discount, total.
5	Payment Portal	Secure input for Cash, Card, or UPI payment method.
6	Receipt Display	Final formatted receipt printed to console and saved to file.

9. Results and Discussion

The Billing and Receipt System was successfully implemented and tested across multiple transaction scenarios. The results demonstrate that:

- The system effectively automates invoicing and receipt processing, reducing the scope for manual arithmetic errors to zero.
- Payment turnaround time improved significantly compared to manual billing — receipt generation is instantaneous after payment confirmation.
- Customer satisfaction is enhanced by providing a clear, itemised, and timestamped receipt for every transaction.
- The tax and discount logic correctly distinguishes between high-value and small-purchase transactions and applies the appropriate financial rule.
- The file-saving feature (receipt.txt) ensures that transaction records persist beyond the program session, supporting basic bookkeeping requirements.
- All three payment modes — Cash (with change calculation), Card, and UPI — were tested successfully.

One known limitation is that the system does not yet persist a full transaction history across multiple sessions. Each run of the program overwrites receipt.txt with the latest receipt. This is identified as a priority enhancement for the next version.

10. Conclusion and Future Enhancements

10.1 Conclusion

The Billing and Receipt System developed in C provides a robust, lightweight platform for handling common business financial transactions through a structured console interface. The project successfully demonstrates practical application of key C programming concepts including structs, arrays, functions, file I/O, and control flow. It fulfills all stated objectives and serves as a strong foundation for a full-featured billing application.

10.2 Future Enhancements

The following improvements are proposed for future iterations of the system:

Enhancement	Description
AI-based Fraud Detection	Automatically flag suspicious transactions using anomaly detection.
Blockchain Logging	Use distributed ledger for tamper-proof transaction records.
Advanced Analytics	Dashboard showing revenue trends and customer payment behaviour.
Mobile App Integration	Android/iOS interface for billing on the go.
Multi-language Support	Interface available in Tamil, Hindi, and other regional languages.
Database Backend	Replace file I/O with MySQL for multi-session transaction history.
Email/SMS Receipts	Automatically send digital receipts to customer's registered contact.
Inventory Management	Real-time stock update and low-stock alerts integrated into billing.

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